substituting or shuffling the letters from the alphabet. It's time we now move on to the digital age where machines do the difficult calculations required to encrypt and decrypt messages, while the work of us mortals has been reduced to figuring out the techniques which would produce stronger ciphers. One of the biggest differences between traditional or classical ciphers, and modern ciphers, lies in the smallest unit which they can operate on. Classical methods had a single letter as the smallest unit, while the modern methods have a single bit as the smallest operable unit.

As machines evolved and the need to communicate between machines grew, standardisation of protocols could not have been avoided for too long. ASCII emerged as the most popular text encoding technique. It worked by treating every number as an integer (and that is a type of simple coding technique again, but was meant to be open) and the integers in turn could be represented as a sequence of bits – hence the modern techniques (which are mostly developed for computers) can work with bits.

Modern techniques of ciphering and deciphering heavily depend on numbers (all text is actually numbers in ASCII, remember?) and can be classified into two broad categories: symmetric and asymmetric.

Symmetric algorithms: A method in which the same key is used for encrypting as well as decrypting the text is called symmetric algorithm. All the classical ciphers we've illustrated thus far would fall under this category because you would always use the same key to encrypt as well as decrypt the message. Some of the better known examples of symmetric algorithms are DES, 3DES (or Triple DES), AES and Blowfish.

Asymmetric algorithms: A method in which the key used for encrypting the message cannot be used to decrypt the message. The key which is to be used for decrypting the message has to be different from the key used to encrypt the message and the relationship between the two keys is complex enough that the second key cannot be derived from the first key. The most popular example of this breed is 'RSA'. If you ever check the SSL certificates of websites you visit, chances are you've read it, unless of course you already know about RSA.

Asymmetric algorithms gain importance over symmetric algorithms because of the fact that symmetric algorithms depend on the same key for encryption as well as for decryption. Since the key is also to be transmitted to the receiver, any person who is able to intercept the encrypted message as well as the key would be able to decrypt the message, thus rendering encryption completely useless. Since with Asymmetric algo-